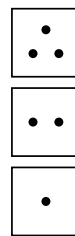
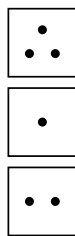
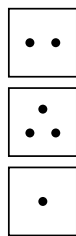
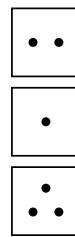
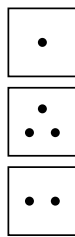
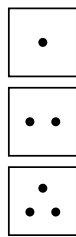
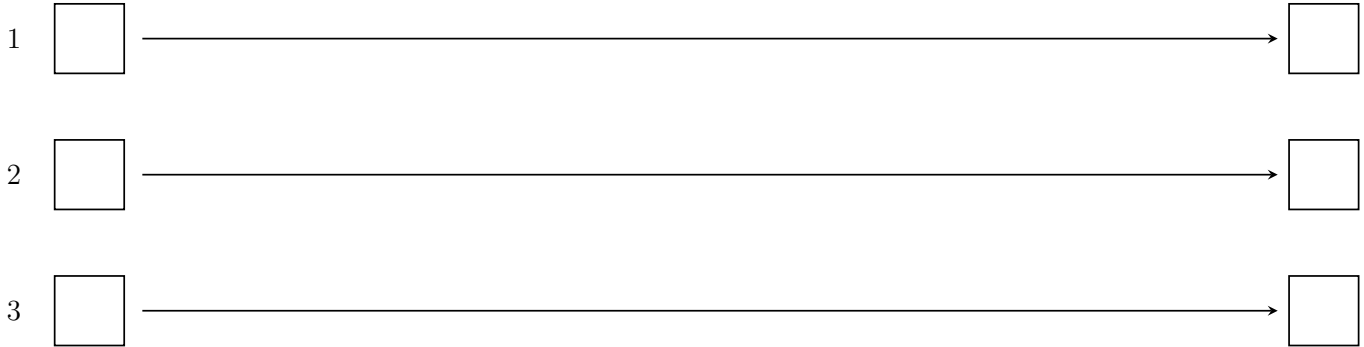
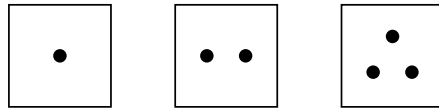


Move left to right. A bar compares the two lanes with black dots: fewer dots go above, more dots below. Equal cards may stay put. Keep every bar fixed and move cards only at bars. A machine sorts when the cards finish in order from fewest dots at the top to most dots at the bottom.

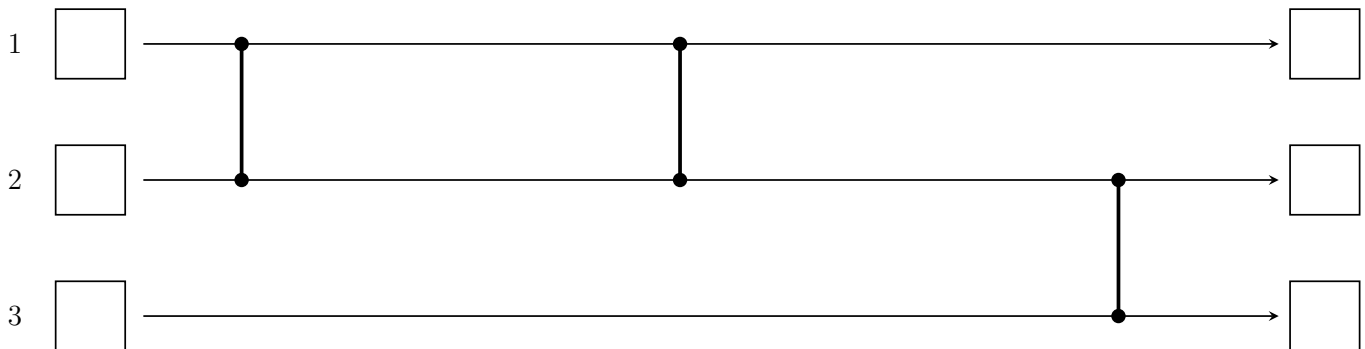
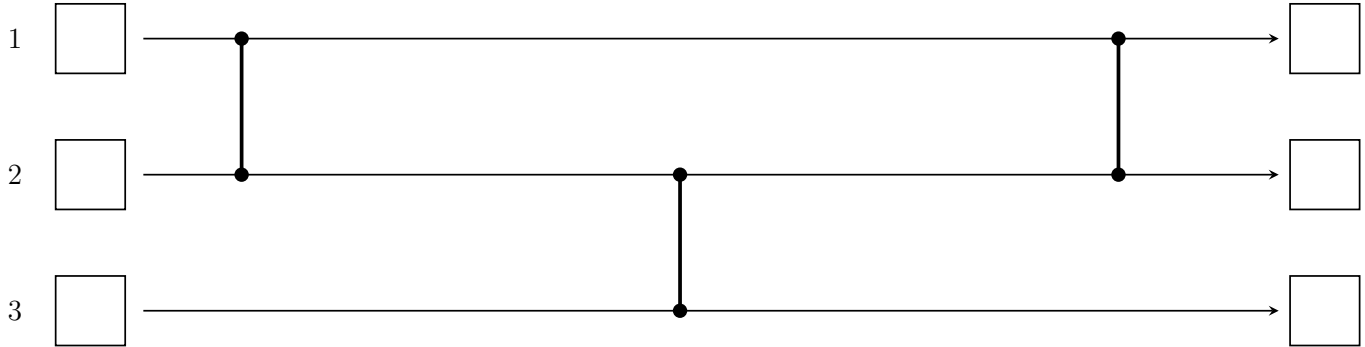
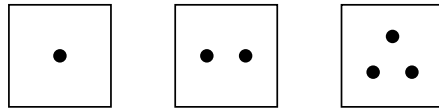
Problem 1: Try every start. Circle the starts that finish in order.



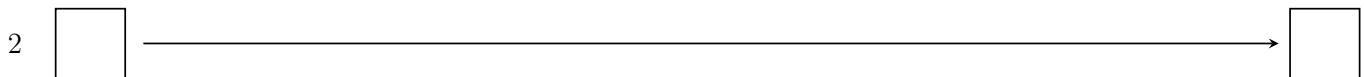
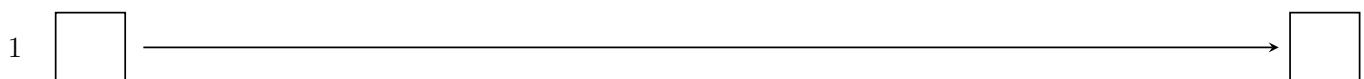
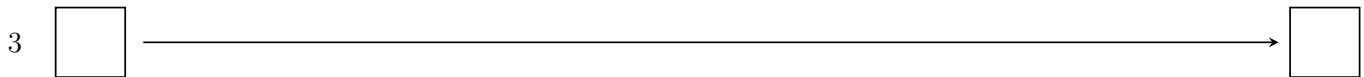
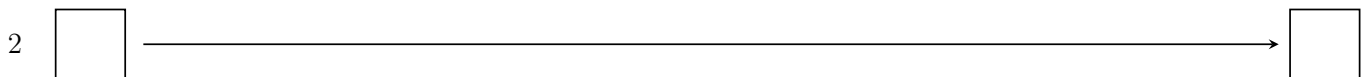
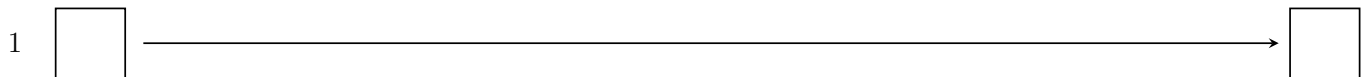
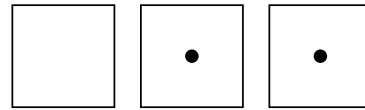
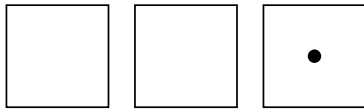
Problem 2: Build a machine that sorts these three cards from every start.
Use as few bars as you can.



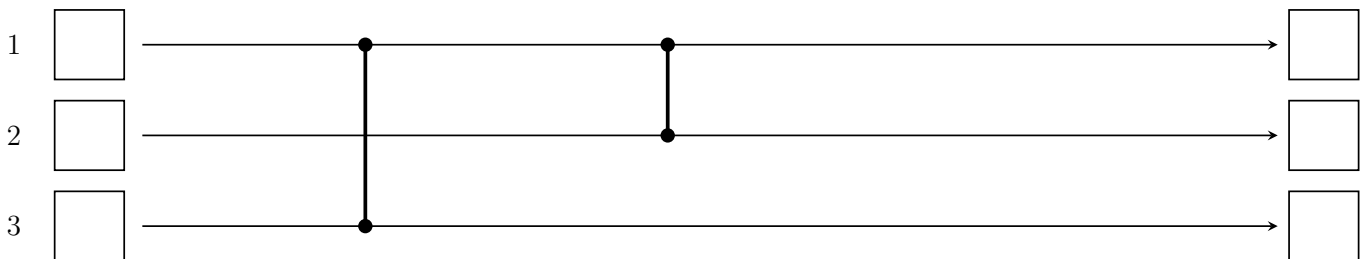
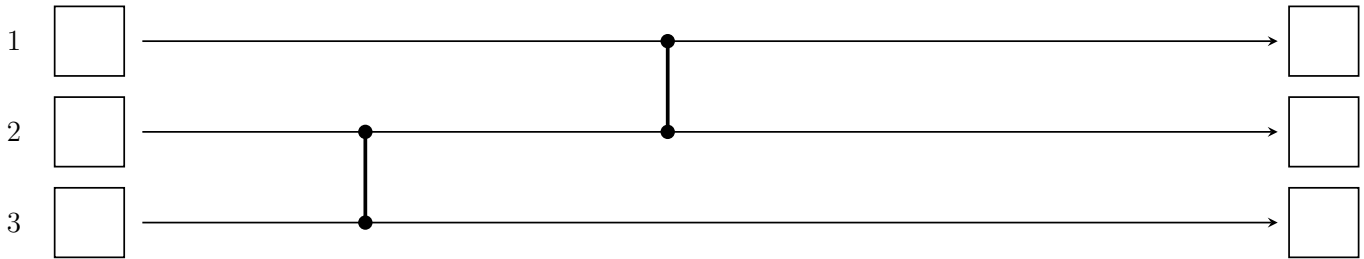
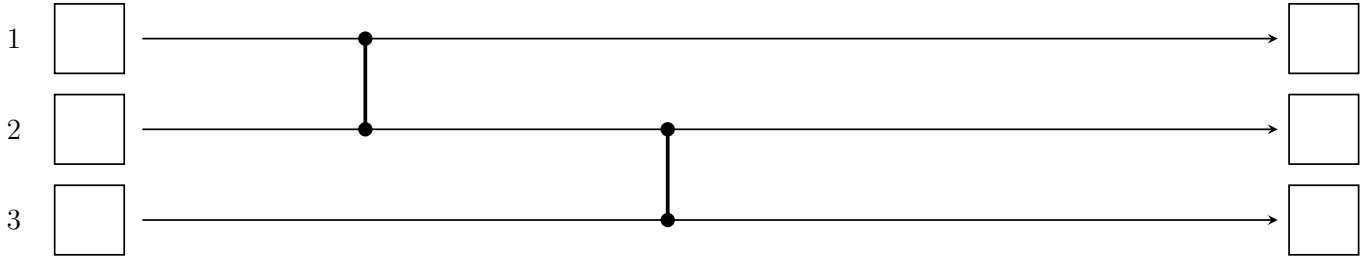
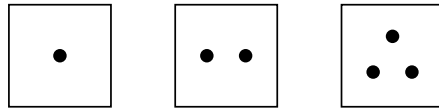
Problem 3: Use the three cards. Find every start that makes either machine finish in the wrong order.



Problem 4: Build a machine that always sorts the left set but sometimes fails on the right set. Make another that always sorts the right set but sometimes fails on the left set.



Problem 5: Each machine needs one more bar at the right. Add it so that every start of the three cards finishes in order.



Problem 6: Build a machine that sorts these four cards from every start. Use as few bars as you can.

